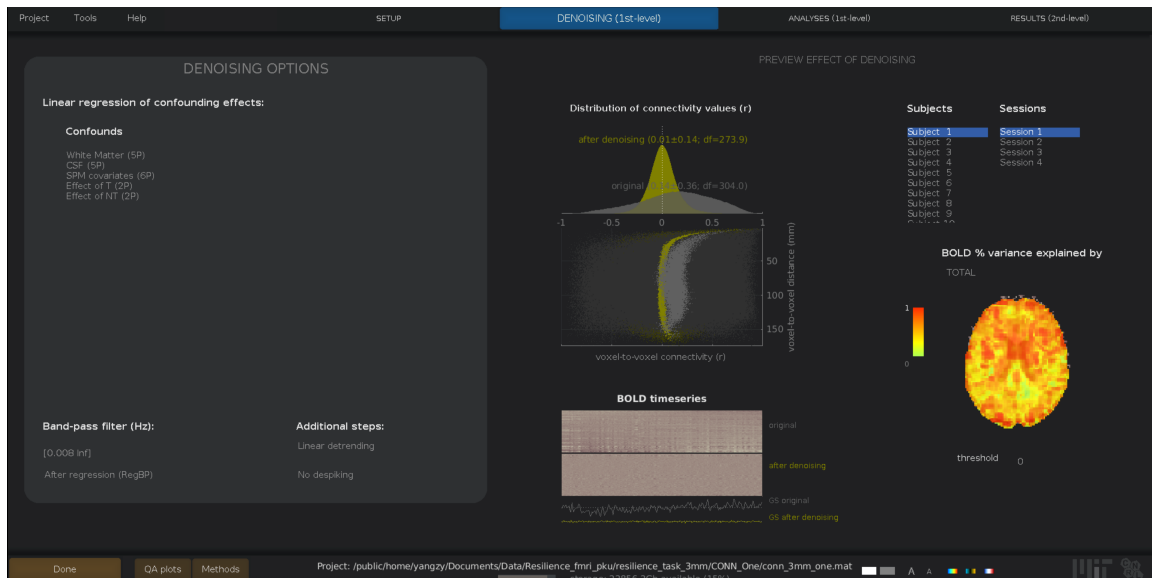


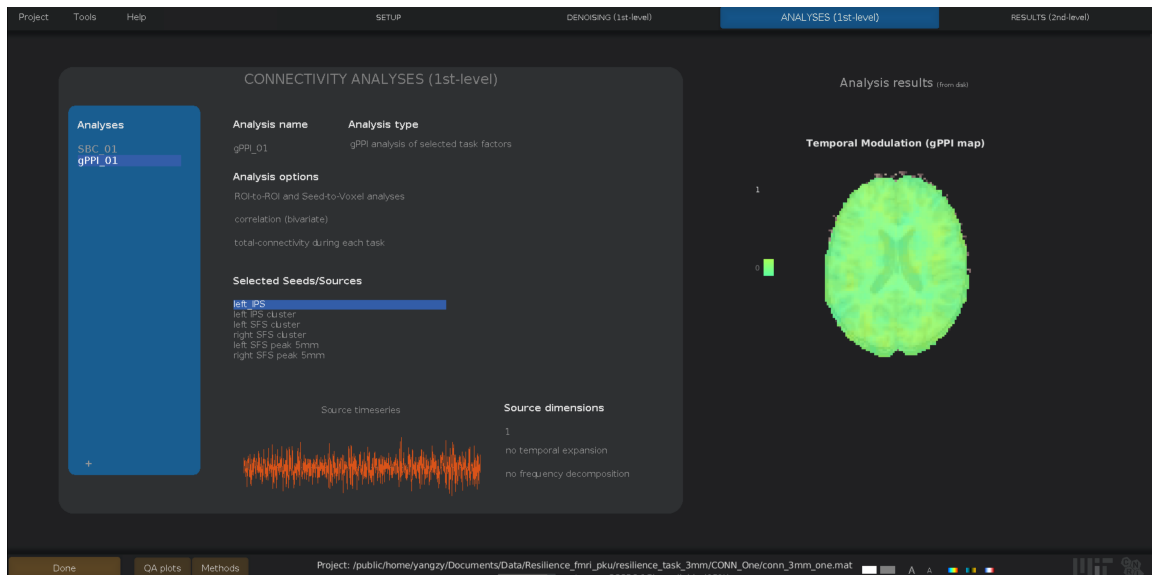
To ensure BOLD signal purity before connectivity estimation, the following confounds were addressed in the Denoising pipeline:



- **Confound Regression (aCompCor):** Regressed out principal components from White Matter (WM) and Cerebrospinal Fluid (CSF) to control for physiological noise.
- **Motion & Scrubbing:** Regressed out subject-specific realignment parameters (and their first-order derivatives) alongside scrubbing covariates to mitigate motion artifacts.
- **Temporal Filtering:** Applied a standard band-pass filter to restrict the BOLD signal to low-frequency fluctuations of interest.

3. First-Level Analysis (Analyses Tab)

The gPPI model was specifically constructed at the first level to evaluate task-modulated connectivity:

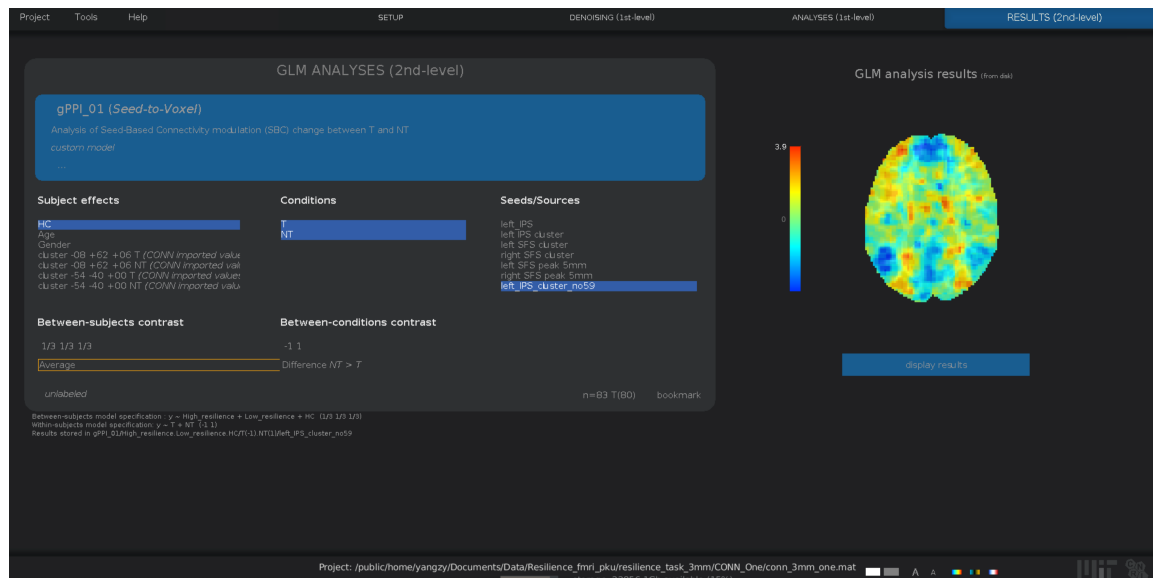


- **Analysis Type:** Selected "Task-modulation effects (gPPI)".

- **Model Design:** For each predefined ROI, the gPPI model was set to include the seed time series, the task condition regressors, and their interaction terms.
- **Metric Quantification:** Connectivity changes for the specific task conditions were quantified using Fisher-transformed correlation coefficients (i.e., the standardized regression coefficient) derived from the interaction terms in the gPPI model.

4. Second-Level Analysis (Results Tab)

Group-level inferences were computed to establish statistical significance:



- **Group-level Design Matrix:** The model included three distinct groups to facilitate both within-group and between-group comparisons.
- **Main Effect of Task (Across Groups):** We first evaluated the overall "No-Think vs. Think" contrast (NT > T; weighted [1 -1]) collapsed across all three groups. This analysis aimed to identify brain regions showing consistent task-modulated connectivity changes regardless of group membership.
- **Between-Group Differences (F-test):** To investigate whether the suppression-related connectivity (NT - T) differed significantly between the three groups, an F-test was performed. This test assessed the "Group × Condition" interaction by testing the null hypothesis that the "No-Think vs. Think" effect was equivalent across all groups.
- **Statistical Thresholding:** Results were thresholded using a primary voxel-level $p < 0.001$ (uncorrected), followed by a cluster-level False Discovery Rate (FDR) correction at $pFDR < 0.05$.